

INVESTIGATION REPORT

Issue 1017: Result memorisation enable in the EX stage

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1017
Disposition	Dead code, proven
Record date	2026-09-17
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1017
Issue filed	2024-07-02
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL 6033d2b, COREV_PULP=1 FPU=1 (the upstream PULP_F0 configuration), top cv32e40p_formal_top, yosys-slang frontend
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	4 cited, 4 present

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The uncovered expression row of the if at cv32e40p_ex_stage.sv:387 is unreachable. Line 387 is the enable condition of the APU_Result_Memorization register: it latches an APU result that arrives while the write-back port is busy. The row never exercised by regression is the one reached with mulh_active set while mult_operator_i is not MUL_H, with every other disjunct of the condition false. mulh_active is only raised by the multiplier for a MUL_H operation, so that combination cannot occur. An unbounded proof (abc pdr, top cv32e40p_formal_top, COREV_PULP=1 FPU=1) passes, and a bounded cover run reaches the full antecedent at step 14, so the assertion is not vacuous. This is the same property as the upstream assert_unreachable_ex_387, checked with a different tool chain.

3 Conditions

1. legal OBI handshakes on both the instruction and the data interface
2. scan_cg_en_i=0
3. no CSR write to the hardware-loop CSRs (0xCC0, 0xCC1, 0xCC2, 0xCC4, 0xCC5, 0xCC6)
4. clock gates modelled as transparent
5. every trace starts in reset
6. instruction memory contents otherwise unconstrained

Reproduced from the assumes that are enabled in the upstream formal environment at cores/cv32e40p/scripts/formal/src, plus the two modelling choices the open-source flow requires.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_ex_stage.sv:387 (enable of APU_Result_Memorization): the expression row with mulh_active=1 and mult_operator_i != MUL_H, all other disjuncts false, is formally proven unreachable (abc pdr, unbounded, top cv32e40p_formal_top, COREV_PULP=1 FPU=1, RTL 6033d2b). mulh_active is raised only for MUL_H. Assumptions: legal OBI handshakes on both interfaces, scan_cg_en_i=0, no CSR write to the hardware-loop CSRs, transparent clock gates. Evidence: runs/v1-cv32e40p-1017/formal/p_assert/sby.log (PASS) and formal/c_antecedent/sby.log (antecedent reached at step 14). Exclude that row only.

Checker used

checker_1017.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (COREV_PULP=1 FPU=1): whenever the line 387 antecedent holds, mult_operator_i == MUL_H
E2	proof	formal/p_assert/result.json	present	machine-readable status record for the proof job
E3	proof	formal/c_antecedent/sby.log	present	smtbmc/yices cover PASS (depth 60): the full antecedent is reached at step 14, so a_ex387 is not vacuous
E4	proof	formal/c_antecedent/result.json	present	machine-readable status record for the cover job

Paths are relative to runs/v1-cv32e40p-1017. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

7 Limitations and scope

- The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/v1-cv32e40p-1017
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024: waived in v2; reported proven unreachable with Siemens Questa Static formal, assertion assert_unreachable_ex_387
Assessed confidence	0.9. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

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